

ETX1100/ETX5900 Business Statistics
Week 8: Revision Workshop
Workshop Lead: Alyssa Hu

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Figure 1: ETX5900 source PDF, Week 8, page 1

Exam Format

- Online supervised eExam
- Closed book exam with authorised materials (see next few slides)
- Exam timetable: 11 February at 10AM
- Time limit: 2 hours and 10 minutes
- Attempt all questions. The exam is marked out of a total of 50 marks.
- The exam is 50% of the unit's assessment value.



Important:

It is your responsibility to be prepared for the exam conditions. Please read this page carefully to understand what an eExam entails: https://www.monash.edu/students/admin/assessments/exams/off-campus#tabs__2545550

Figure 2: ETX5900 source PDF, Week 8, page 2

Exam Format

Part A (20 marks)

- **ETX1100:** 20 MCQs
- **ETX5900:** 10 MCQs & 10 numerical questions

Part B (30 marks)

3 short answer questions (10 marks each)

- Write your answers on paper
- Show all working on paper
- You are given time to upload answers using a QR code (30 minutes to upload - after the exam is over)

Figure 3: ETX5900 source PDF, Week 8, page 3

Authorised Materials

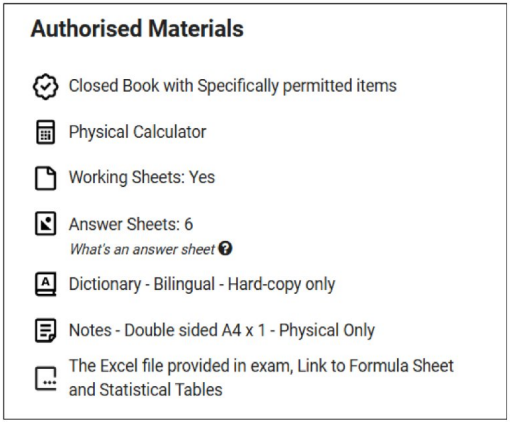


Figure 4: ETX5900 source PDF, Week 8, page 4

Authorised Materials

Notes

You are allowed **ONE** printed or hand-written **A4 sheet** of **double-sided** notes. It must be a **physical** sheet, not a computer file.

Excel file:

- An empty Excel file with Monash watermarks will be provided within the exam link on the day of the exam.
- Download the provided Excel file to help with the calculations.
- You cannot have any opened Excel files when starting the exam OR open any other Excel files during the exam.

Figure 5: ETX5900 source PDF, Week 8, page 5

Materials Not Allowed

Reminders:

- Soft copy files are not allowed.
- You are not allowed to use a second device.
- You are not allowed to log into any website or engage in any online communication.
- You are not allowed to open any Excel file other than the one provided in the exam.
- You are not allowed to use any online translation software.
- You are not allowed to use AI.

Figure 6: ETX5900 source PDF, Week 8, page 6

Exam Preparation

✓There is **NO** hurdle requirement for this unit.

Refer to the following for revision:

✓Workshop material (pre-recorded and live) and all quizzes

✓Textbook questions as listed on Moodle. Solutions are provided.

✓Mock exam and past exams



Figure 7: ETX5900 source PDF, Week 8, page 7

Material Covered in the Final Exam

Weeks 2 – 8

Week 2: Basic Probability and Introduction to Normal Distribution
Week 3: Sampling Distribution and Confidence Interval Estimation
Week 4: Hypothesis Testing I: Chi-Square Test of Independence
Week 5: Hypothesis Testing II: Population Mean and Proportion
Week 6: Correlation and Regression Analysis I
Week 7: Correlation & Regression Analysis II; Time Series Analysis & Forecasting
Week 8: Revision

Figure 8: ETX5900 source PDF, Week 8, page 8

Pre-Exam Consultation and Announcements

- Check Moodle regularly for important announcements
- Please check your marks on the grade book carefully
- Pre-exam consultation times to be posted on Moodle

All the best for the exam!



Figure 9: ETX5900 source PDF, Week 8, page 9

Practice Questions

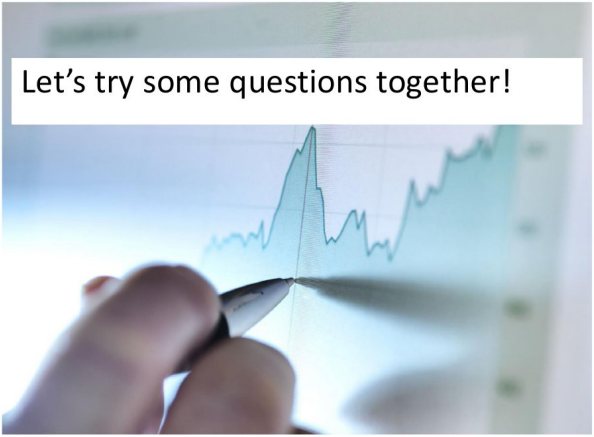




Figure 10: ETX5900 source PDF, Week 8, page 10

Question 1

The sales manager of a clothing company claims that the mean spending by customers is less than $80. A sample of 50 customers has a mean spending of $76.25 and a standard deviation of $15.25. We wish to perform an appropriate hypothesis test at the 5% level of significance.

a) State the hypotheses

b) What is the p-value?

c) Interpret the results in the context of the question.

d) Is there evidence to support the manager's claim?

e) What is a potential error in the test above?

Figure 11: ETX5900 source PDF, Week 8, page 11

Question 1_solution

The sales manager of a clothing company reports that the mean spending by customers is less than $80. A sample of 50 customers has a mean spending of $76.25 and a standard deviation of $15.25. We wish to perform an appropriate hypothesis test at the 5% level of significance. **(Note: Include all steps to get full marks.)**

a) State the hypotheses

H_0: $\mu = 80$

H_1: $\mu < 80$

b) What is the p-value?

Test statistic = $\frac{(76.25-80)}{15.25/\sqrt{50}} = -1.739$

P-value = P(t<-1.739) = T.DIST(-1.739,49,TRUE) =0.044176

Figure 12: ETX5900 source PDF, Week 8, page 12

Question 1_solution

c) Interpret the results in the context of the question.

We reject the null as the p-value is less than 0.05. There is sufficient evidence at the 5% level of significance that the true mean spending is less than $80.

d) Is there evidence to support the manager's claim?

Yes, as per our result in part c, there is evidence to support the manager's claim.

e) What is a potential error in the test above? Interpret the error in the context of the question.

Type 1 error where H_0 is true and we reject it.

We conclude that the true mean spending is less than $80 when it actually is not.

Figure 13: ETX5900 source PDF, Week 8, page 13

Question 2

A market researcher states that she is 95% confident that the true mean monthly sales of a product are between $180,000 and $200,000. Explain the meaning of this statement.

Figure 14: ETX5900 source PDF, Week 8, page 14

Question 2_solution

A market researcher states that she is 95% confident that the true mean monthly sales of a product are between $180,000 and $200,000. Explain the meaning of this statement.

If all possible samples of the same size n are taken,

and we calculate the confidence intervals,

95% of the intervals include the true population average monthly sales of the product within the interval developed.

Figure 15: ETX5900 source PDF, Week 8, page 15

Question 3

The Excel output below contains observations on variables related to the sales price ($) of houses in Shallowhaven. The variables are house size (square feet) and number of bedrooms.

a. State the estimated multiple regression equation. Define any symbols used.
b. Interpret the meaning of the slopes in the estimated equation.
c. At the 5% level of significance, test if there is a significant positive linear relationship between house price and number of bedrooms.

Regression Statistics						
Multiple R	0.723195					
R Square	0.523011					
Adjusted R Square	0.516522					
Standard Error	28933.13					
Observations	150					
ANOVA						
	df	SS	MS	F	Significance F	
Regression	2	1.34931E+11	67465519362	80.5918	2.34595E-24	
Residual	147	1.23058E+11	837126119.5			
Total	149	2.57989E+11				
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	59884.907	7812.5844	7.6651853	2.23E-12	44445.418	75324.3970
House_Size	32.778057	6.3373394	5.172211	7.45E-07	20.253996	45.302118
Number_Bedrooms	17834.123	5250.0616	3.3969359	0.000877	7458.776	28209.469

Figure 16: ETX5900 source PDF, Week 8, page 16

Question 3_solutions

a) State the estimated multiple regression equation. Define any symbols used.

$$\hat{Y} = 59884.907 + 32.778057X_1 + 17834.123X_2$$

where X_1 is house size (square feet), X_2 is number of bedrooms and $\hat{y}$ is the estimated sales price of houses ($)

b) Interpret the meaning of the slopes in the estimated equation.

b_1 = 32.778057: On average, sales prices of houses are expected to increase by $32.78 for each extra square foot in size, holding constant the number of bedrooms.

b_2 = 17834.123: On average, sales prices of houses are expected to increase by $17,834.12 for each extra bedroom, holding constant the house size.

Figure 17: ETX5900 source PDF, Week 8, page 17

Question 3_solutions

c) At the 5% level of significance, test if there is a significant positive linear relationship between house price ($) and number of bedrooms.

Step 1:

$H_0: \beta_2 = 0$

$H_1: \beta_2 > 0$

Step 2:

Level of Significance = 0.05

t-statistic = 3.397

Step 3: *p-value* = 0.000877/2 ≈ 0

Step 4:

Reject H_0 if p-value < α.

Since 0 < 0.05, reject H_0.

Regression Statistics						
Multiple R	0.723195					
R Square	0.523011					
Adjusted R Square	0.516522					
Standard Error	28933.13					
Observations	150					
ANOVA						
	df	SS	MS	F	Significance F	
Regression	2	1.34931E+11	67465519362	80.5918	2.34595E-24	
Residual	147	1.23058E+11	837126119.5			
Total	149	2.57989E+11				
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	59884.907	7812.5844	7.6651853	2.23E-12	44445.418	75324.3970
House_Size	32.778057	6.3373394	5.172211	7.45E-07	20.253996	45.302118
Number_Bedrooms	17834.123	5250.0616	3.3969359	0.000877	7458.776	28209.469

Step 5:

There is sufficient evidence at the 5% level of significance to conclude that there is a positive linear relationship between sales price of houses ($) and the number of bedrooms.

Figure 18: ETX5900 source PDF, Week 8, page 18

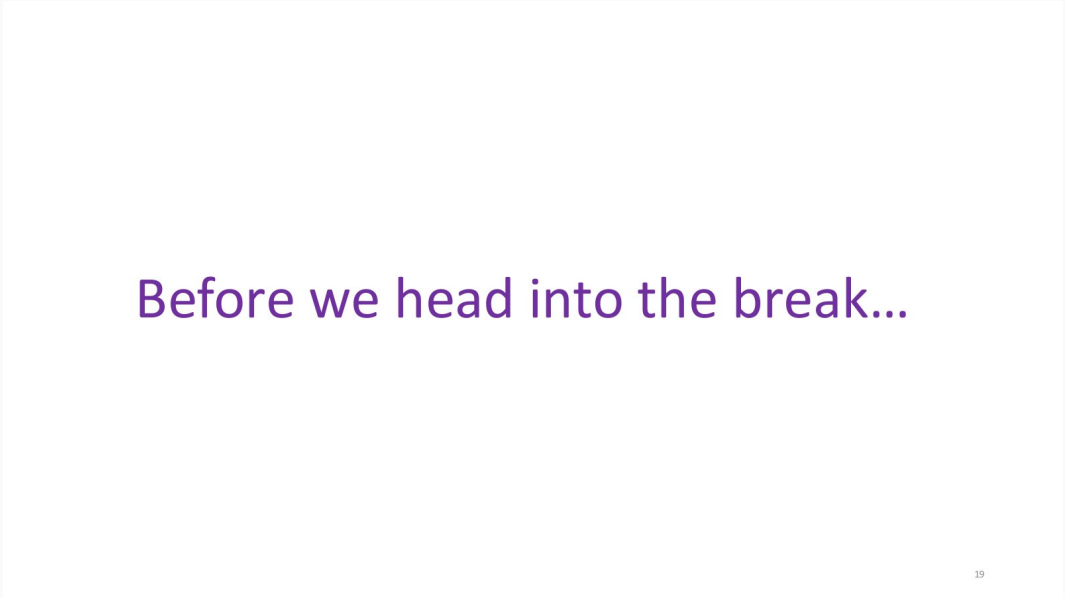


Figure 19: ETX5900 source PDF, Week 8, page 19

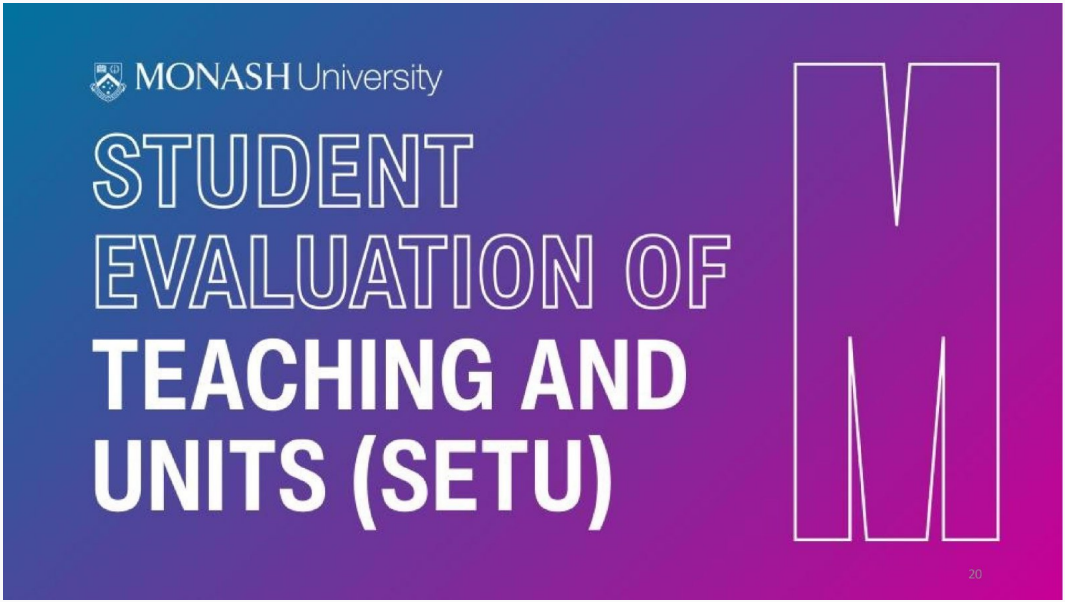


Figure 20: ETX5900 source PDF, Week 8, page 20

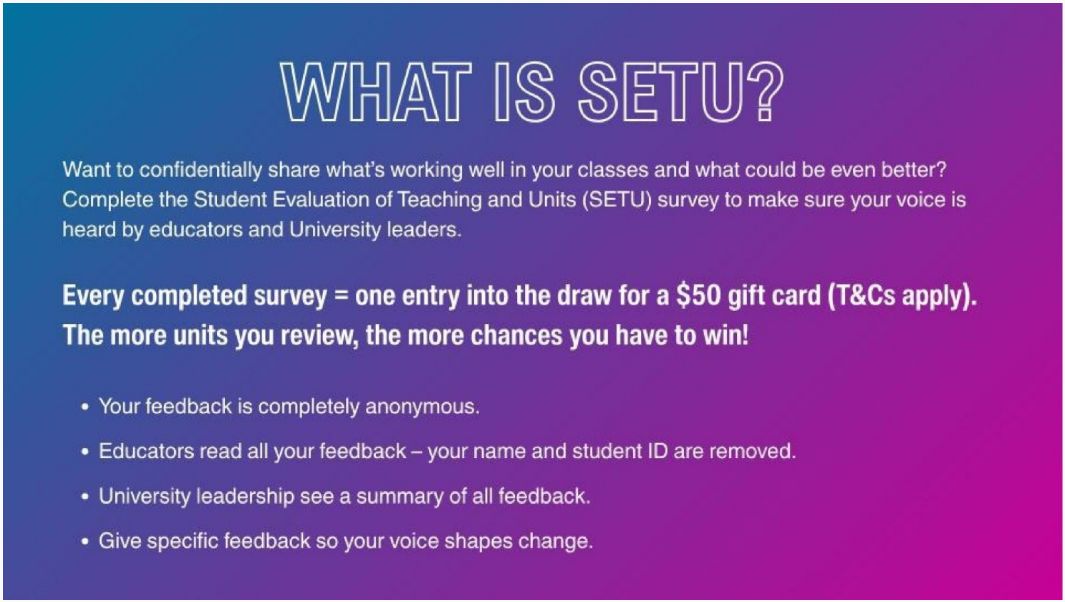


Figure 21: ETX5900 source PDF, Week 8, page 21



Figure 22: ETX5900 source PDF, Week 8, page 22

Question 4

A popular weekly magazine asserts that fewer than 40% of households in Australia claim to have changed their lifestyles because of escalating energy prices.

Recently, a survey of 180 households was conducted including this question, and 63 of these households claimed to have made lifestyle changes due to escalating energy prices.

a) Assuming the data obtained is a random sample, is there evidence to support the magazine's claim at the 10% level of significance. Use the p-value method.

b) Would the result of the test have been different if you had used a 5% significance level? Explain.

c) Obtain a 90% confidence interval for the true proportion of Australian households that claim to have made lifestyle changes due to escalating energy prices.

Figure 23: ETX5900 source PDF, Week 8, page 23

Question 4_solutions

a) Assuming the data obtained is a random sample, is there evidence to support the magazine's claim at the 10% level of significance. Use the p-value method.

Let X = number of households that made lifestyle change due to increasing energy prices

Rule of thumb check: $n\pi = 180 \times 0.40 = 72$ and $n(1-\pi) = 80(0.60) = 108$ are both > 5

The normal approximation is reasonable.

Step 1: Hypothesis:

H_0: $\pi = 0.40$ (or $\pi \geq 0.40$)

H_1: $\pi < 0.40$

$n=180$; $X=63$; $p=63/180 = 0.35$; $\alpha=0.10$

Step 2: Test Statistic:

For π always use Z

$$Z = \frac{p - \pi_0}{\sqrt{\frac{\pi_0(1-\pi_0)}{n}}} \qquad Z_{calc} = \frac{0.35-0.40}{\sqrt{\frac{0.40(1-0.40)}{180}}} = -1.369 \approx -1.37$$

Figure 24: ETX5900 source PDF, Week 8, page 24

Question 4_solutions

Step 3: p – value

So, P(Z<-1.37)

=0.0853 **=NORM.S.DIST(-1.37,true)**

Step 4: Decision Rule

Reject H_0 if p-value < α;

Since 0.0853 < 0.10, we can reject H_0.

Step 5: Conclusion

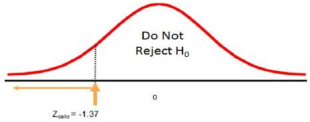
There is sufficient evidence at the 10% level of significance that the true proportion of households who have made the change to lifestyle due to increasing energy prices **is** less than 40%.

The data supports the magazine's claim.

Figure 25: ETX5900 source PDF, Week 8, page 25

Question 4_solutions

b) Would the result of the test have been different if you had used a 5% significance level? Explain.



Yes.

Since 0.0853 > 0.05, we CANNOT reject H_0

There is insufficient evidence at the 5% level of significance that the true proportion of households who have made the change to lifestyle due to increasing energy prices **is** less than 40%.

There is no evidence to support the magazine's claim at the 5% significance level.

Figure 26: ETX5900 source PDF, Week 8, page 26

Question 4_solutions

c) Obtain a 90% confidence interval for the true proportion of Australian households that claim to have made lifestyle changes due to escalating energy prices.

Since np=180*0.35=63 and n(1-p)=180(1-0.35) = 117 are both >5, the normal approximation is reasonable.

$$p - Z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}} < \pi < p + Z_{\alpha/2}\sqrt{\frac{p(1-p)}{n}}$$

$$0.35 - 1.645\sqrt{\frac{0.35(1-0.35)}{180}} < \pi < 0.35 + 1.645\sqrt{\frac{0.35(1-0.35)}{180}}$$

$$0.35 - 1.645 * 0.03555 < \pi < 0.35 + 1.645 * 0.03555$$

$$0.2915 < \pi < 0.40848$$

We are 90% confident that the true proportion of households who have made the change to lifestyle due to increasing energy prices is estimated to lie, on average between 29.15% and 40.85%

Figure 27: ETX5900 source PDF, Week 8, page 27

Quiz Question 5

A manufacturer of energy-efficient washing machines advertises that their machines use an average of 112 litres of water per cycle. Some environmental groups are concerned that the machines may, in practice, use more water than advertised.

To investigate, a consumer group randomly selects 16 households using this washing machine model and records water usage per cycle. The sample mean is 115.3 litres, with a sample standard deviation of 6.4 litres. Assume the water usage per cycle is approximately normally distributed. Perform an appropriate hypothesis test at the 5% level of significance using the p-value method.

State the hypotheses?

A. H_0: $\mu = 112$; H_1: $\mu > 112$
B. H_0: $\mu = 112$; H_1: $\mu < 112$
C. H_0: $\pi = 112$; H_1: $\pi > 112$
D. H_0: $\pi = 112$; H_1: $\pi < 112$

Figure 28: ETX5900 source PDF, Week 8, page 28

Quiz -Question 5

A manufacturer of energy-efficient washing machines advertises that their machines use an average of 112 litres of water per cycle. Some environmental groups are concerned that the machines may, in practice, use more water than advertised.

To investigate, a consumer group randomly selects 16 households using this washing machine model and records water usage per cycle. The sample mean is 115.3 litres, with a sample standard deviation of 6.4 litres. Assume the water usage per cycle is approximately normally distributed. Perform an appropriate hypothesis test at the 5% level of significance using the p-value method.

State the hypotheses?

A. H_0: $\mu=112$; H_1: $\mu>112$
B. H_0: $\mu=112$; H_1: $\mu<112$
C. H_0: $\pi=112$; H_1: $\pi>112$
D. H_0: $\pi=112$; H_1: $\pi<112$

Figure 29: ETX5900 source PDF, Week 8, page 29

Quiz Question 6

A manufacturer of energy-efficient washing machines advertises that their machines use an average of 112 litres of water per cycle. Some environmental groups are concerned that the machines may, in practice, use more water than advertised.

To investigate, a consumer group randomly selects 16 households using this washing machine model and records water usage per cycle. The sample mean is 115.3 litres, with a sample standard deviation of 6.4 litres. Assume the water usage per cycle is approximately normally distributed. Perform an appropriate hypothesis test at the 5% level of significance using the p-value method. State your answer to 4 decimals.

The test statistic is _____ and the p-value is _____

Figure 30: ETX5900 source PDF, Week 8, page 30

Quiz –Question 6

A manufacturer of energy-efficient washing machines advertises that their machines use an average of 112 litres of water per cycle. Some environmental groups are concerned that the machines may, in practice, use more water than advertised.

To investigate, a consumer group randomly selects 16 households using this washing machine model and records water usage per cycle. The sample mean is 115.3 litres, with a sample standard deviation of 6.4 litres. Assume the water usage per cycle is approximately normally distributed. Perform an appropriate hypothesis test at the 5% level of significance using the p-value method. State your answer to 4 decimals.

The test statistic is 2.0625

and the p-value is 0.0285

Question 2		Formula
mu	112	
Xbar	115.3	
s	6.4	
alpha	0.05	
n	16	
Test Statistic	2.06250	(xbar -mu)/ (s/sqrt(n))
P-value	0.028461	T.DIST.RT(test stat, df = n-1)

Can also use 1-T.DIST(test stat,df,true)

Figure 31: ETX5900 source PDF, Week 8, page 31

Quiz Question 7

An electric car manufacturer claims that their new battery model increases average driving range by more than 5 km after a full overnight charge. To evaluate this claim, an independent reviewer tests the battery on 17 vehicles. The sample shows an average increase of 5.8 km, with a standard deviation of 1.2 km. Assume the increase in range is approximately normally distributed. Use a significance level of 0.05.

If the test statistic is 2.582, which of the following is the correct decision and conclusion regarding the manufacturer's claim?

A. Fail to reject the null hypothesis; there is not enough evidence to support the manufacturer's claim.
B. Reject the null hypothesis; the average range increase is significantly less than 5 km.
C. Reject the null hypothesis; there is sufficient evidence that the battery increases range by more than 5 km.
D. Fail to reject the null hypothesis; the average range increase is exactly 5 km.

Figure 32: ETX5900 source PDF, Week 8, page 32

Quiz –Question 7

An electric car manufacturer claims that their new battery model increases average driving range by more than 5 km after a full overnight charge. To evaluate this claim, an independent reviewer tests the battery on 17 vehicles. The sample shows an average increase of 5.8 km, with a standard deviation of 1.2 km. Assume the increase in range is approximately normally distributed. Use a significance level of 0.05.

If the test statistic is 2.582, which of the following is the correct decision and conclusion regarding the manufacturer's claim?

n	17	
Test Stat	2.582	
P-value	0.01003	T.DIST.RT(test stat, df = n-1)

Can also use 1-T.DIST(test stat,df,true)

A. Fail to reject the null hypothesis; there is not enough evidence to support the manufacturer's claim.

B. Reject the null hypothesis; the average range increase is significantly less than 5 km.

C. Reject the null hypothesis; there is sufficient evidence that the battery increases range by more than 5 km.

D. Fail to reject the null hypothesis; the average range increase is exactly 5 km.

Figure 33: ETX5900 source PDF, Week 8, page 33

Quiz Question 8

The amount of time (in minutes) customers spend in a self-service petrol station follows a normal distribution with a mean of 12 minutes and a standard deviation of 3 minutes.

The probability that a randomly selected customer spends more than 16 minutes at the station is _____ (State your final answer to 4 decimal places)

Figure 34: ETX5900 source PDF, Week 8, page 34

Quiz –Question 8

The amount of time (in minutes) customers spend in a self-service petrol station follows a normal distribution with a mean of 12 minutes and a standard deviation of 3 minutes. What is the probability that a randomly selected customer spends more than 16 minutes at the station?

The probability that a randomly selected customer spends more than 16 minutes at the station is 0.0912 (State your final answer to 4 decimal places).

Question 4				
mu	12			
sigma	3			
x	16			
P(X>x)	0.091211	1- NORM.DIST(x, mu, sigma, cumulative)		

Figure 35: ETX5900 source PDF, Week 8, page 35

Quiz Question 9

The amount of time (in minutes) customers spend in a self-service petrol station follows a normal distribution with a mean of 12 minutes and a standard deviation of 3 minutes.

The minimum number of minutes a customer must spend at the station to be in the top 10% of all customers is _____ minutes (State your final answer to 2 decimal places).

Figure 36: ETX5900 source PDF, Week 8, page 36

Quiz –Question 9

The amount of time (in minutes) customers spend in a self-service petrol station follows a normal distribution with a mean of 12 minutes and a standard deviation of 3 minutes. What is the minimum number of minutes a customer must spend at the station to be in the top 10% of all customers?

The minimum number of minutes a customer must spend at the station to be in the top 10% of all customers is 15.84 minutes (State your final answer to 2 decimal places).

Question 5		
mu	12	
sigma	3	
P(X>x*)	0.1	Need to enter 0.9 as 'probability'!
X	15.84465	NORM.INV(probability, mu, sigma)

Figure 37: ETX5900 source PDF, Week 8, page 37

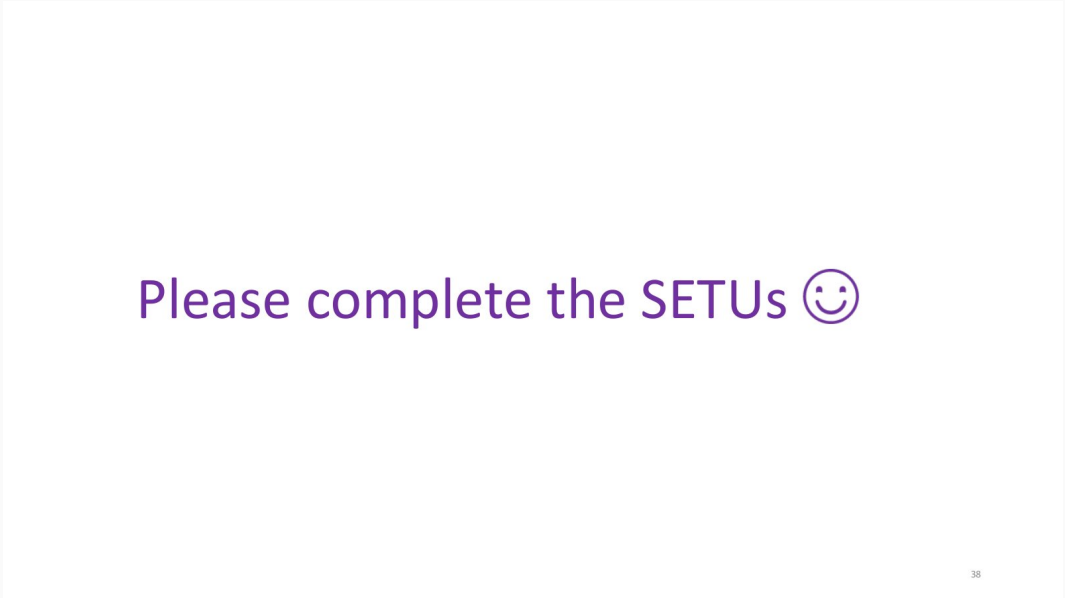


Figure 38: ETX5900 source PDF, Week 8, page 38



Figure 39: ETX5900 source PDF, Week 8, page 39